4. Write the Python program for Crypt - Arithmetic problem

Reg.No: 192511013

Aim

To write a python program to solve the Crypt - Arithmetic problem by assigning unique digits to letters so that the arithmetic equation is correct.

Algorithm

1. Consider the equation SEND + MORE = MONEY
2. Identify all Unique letters in the equation.
3. Assign a different digit to each other.
4. Ensure that the first letter of each word is not zero.
5. Check whether the numerical equation is correct.
6. Display the digit assigned to each letter and the solved equation.

Program

```
from itertools import permutation

letters = "SENDMORY"

for digits in permutation (range (10), len (letters)):
    d = dict (zip ( letters , digits))

    if d ["S"] == 0 or d ["M"] == 0 :
        continue

    Send = 1000*d ["S"] + 100*d ["E"] + 10*d ["N"] + d["D"]

    more = 1000*d ["M"] + 100*d [" O"] + 10*d ["R"] + d["E"]
```

```
money = (10000*d["M"] + 1000*d["O"] +
         100*d["N"] + 10*d["E"] + d["Y"])

If send + more == money:
    print("Solution found:")
    for letter in letters:
        print(letter, "=", d[letter])
    print(send, "+", more, "=", money)
    break
else:
    print("No solution found.")
```

Sample Input

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Sample Output

Solution found:

S = 9

E = 5

N = 6

D = 7

M = 1

O = 0

R = 8

Y = 2

9567 + 1085 = 10652

permutation & checkin

Result

Thus, the Crypt-Arithmetic problem was solved successfully using digit